Perception, Not Reasoning, Limits Visual Theory of Mind

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Abstract

Vision-language models pass text-based false-belief probes at near-human levels but drop by twenty or more percentage points the moment the same problem is conveyed visually. The consensus diagnosis is conceptual, prescribing new belief-reasoning modules. We argue the bottleneck is perceptual: not what the model cannot reason about, but what it cannot see. We propose a five-condition dissociation protocol that holds the reasoning task constant while varying perceptual fidelity from visual only to text only, and quantify the result with the Perceptual Contribution Ratio, a dose-response metric measuring how much of the visual-to-text gap a scaffold of given fidelity closes. Instantiated on abstract 2D panels in the Heider-Simmel tradition and naturalistic 3D game scenes from MuMA-ToM and MindPower across seven VLMs and roughly 147,000 API calls, ground-truth scaffolds close the gap on every regime, matching or exceeding the text-only ceiling on abstract panels and MuMA-ToM. Automatic parsers, including frontier VLMs as zero-shot scene describers, fail. The dominant visual-only error mode is *reality bias* (60% of false-belief failures): models report where an object currently is rather than where an absent agent believes it to be, the signature of a front-end that fails to encode who was present when. The route to better visual social reasoning runs through perception, not new reasoning modules.

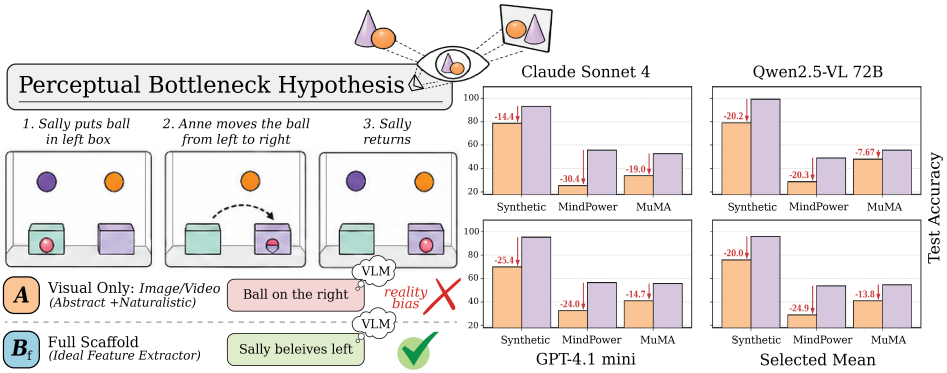


Figure 1: **Visual Theory of Mind is bottlenecked by perception, not reasoning.** *Left:* on a Sally-Anne probe, the VLM under visual-only input (A) reports the ball's current location (*reality bias*); a ground-truth scaffold (B_f) recovers the absent agent's belief without changing the reasoning task. *Right:* per-model visual-only vs. text-only accuracy across abstract 2D panels, MindPower [39], and MuMA-ToM [29]. Ground-truth scaffolds close the gap ($\mathrm{PCR}_f \geq 1$ on abstract and MuMA-ToM); automatic parsers do not.

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40th Conference on Neural Information Processing Systems (NeurIPS 2026). Track on Evaluations and Datasets.

1 Introduction

Children come to read other minds through a sequence of perceptual achievements, not a single conceptual leap. Infants individuate objects behind occluders by three to five months [31], read motion as goal-directed by six to twelve months [38, 9], and pass false-belief tasks by roughly age four [37, 5, 36]. Each capacity requires the previous as input: a goal cannot be attributed to an entity not yet individuated, and a belief cannot be modeled for an agent whose goals cannot be parsed [6, 17]. The Sally-Anne false-belief task, the canonical assay for Theory of Mind, sits at the top of this chain [5, 25].

When this task was adapted for large language models, frontier systems passed text-based probes at near-human levels [16, 33]. The moment the same problem is conveyed visually, performance collapses by twenty or more percentage points across every VLM tested [18, 8], with even larger gaps on naturalistic multi-agent video [29, 14, 39]. The consensus diagnosis is conceptual: VLMs are taken to lack the machinery belief attribution requires, prescribing explicit social-reasoning modules atop existing visual encoders [35, 27, 28].

We argue that this diagnosis is incomplete. The bottleneck is not what the model cannot reason about; it is what it cannot see. If a VLM solves a false-belief problem at 94.5% in text but only 73.9% in pixels, the deficit lies in the perceptual pipeline that converts pixels into propositions, not in the reasoning that operates on them. We call this the *perceptual bottleneck hypothesis*: visual ToM failures reflect breakdowns in structured perceptual-state extraction (entity identity, position, presence, absence across panels or frames), not in downstream belief reasoning. The hypothesis is falsifiable: if perceptual, handing the model the scene description its front-end should have extracted should restore performance toward the text-only ceiling; if conceptual, no scaffold of any fidelity will help.

To test this, we propose a five-condition dissociation protocol that holds the reasoning task constant while varying perceptual fidelity from visual only to text only with three intermediate scaffolds. We instantiate it on two stimulus regimes following prior conventions: *abstract 2D panels* in the Heider-Simmel tradition [11], organized along the developmental cascade, and *naturalistic 3D game scenes* from MuMA-ToM [29] and MindPower [39]. We quantify scaffold benefit with the *Perceptual Contribution Ratio* (PCR), a dose-response metric taking value 1 when a scaffold fully recovers the text-only ceiling and 0 when it does not help.

The signature reproduces across both regimes and seven VLMs spanning three capability tiers. Ground-truth scaffolds close the visual-to-text gap on every regime, matching or exceeding the text-only ceiling on abstract panels and MuMA-ToM. Automatic parsers, including frontier VLMs as zero-shot scene describers, fail. Stripping every explicit visibility statement from the ground-truth scaffold retains most of the gain, ruling out that scaffolds simply hand the model the reasoning predicate. The dominant visual-only error mode (60% of false-belief failures) is what we call *reality bias*: models report where an object currently is rather than where an absent agent believes it to be, the signature of a front-end that fails to encode who was present when. When the front-end cannot deliver entity-level scene states, the inferential pathway downstream cannot compensate; the route to better visual social reasoning runs through perception, not new reasoning modules.

Contributions:

1. A five-condition dissociation protocol isolating perceptual from conceptual contributions to visual ToM failures, applied uniformly across abstract 2D and naturalistic 3D regimes.
2. The Perceptual Contribution Ratio: a dose-response metric tracing how much of the visual-to-text gap a scaffold of given fidelity closes, with negative values flagging misleading scaffolds and values above 1 indicating the combination exceeds the text-only ceiling.
3. Cross-regime evidence, across seven VLMs and roughly $147{,}000$ API calls, that the visual ToM deficit is predominantly perceptual rather than conceptual, with the reality-bias error taxonomy.

2 Related Work

Theory of Mind in language models: Whether LLMs possess Theory of Mind remains contested. Kosinski [16] reported GPT-4 passes false-belief tasks at six-year-old levels; Ullman [35] showed that trivial paraphrases cause catastrophic failures. Strachan *et al*. [33] found above-chance belief tracking with systematic biases absent in humans; Gandhi *et al*. [7] documented uneven performance across ToM dimensions; Shapira *et al*. [28] introduced adversarial controls that degrade LLM ToM; Hu *et*

al. [13] argue existing evaluations conflate linguistic competence with mentalizing; Kim *et al*. [15] stress-tested ToM in information-asymmetric conversations. These studies are text-only. In the visual domain, Jin *et al*. [14] introduced MMToM-QA; Shi *et al*. [29] extended this with MuMA-ToM; Chen *et al*. [8] showed VLMs fail at visual perspective-taking despite intent recognition; Zhang *et al*. [39] documented a substantial human-VLM gap with MindPower. None measures whether the visual deficit is perceptual or conceptual in origin. Our protocol forces that distinction by holding the reasoning task constant and varying only the fidelity of perceptual input, applied uniformly across abstract 2D panels and naturalistic 3D game scenes.

Cognitive benchmarks for VLMs: CogDevelop2K [18] evaluated 46 multimodal LLMs on 12 cognitive sub-concepts organized by developmental milestones, finding models outperform on late-developing abstract reasoning relative to early-developing perceptual tasks. Tong *et al*. [34] traced systematic visual shortcomings to CLIP-based encoders; MotionBench [12] reported substantial gaps in fine-grained video motion understanding. These benchmarks score VLMs but treat the resulting accuracy as a single number; perceptual and inferential contributions remain entangled. Our scaffolding intervention dissociates the two and assigns the deficit to perception rather than reasoning.

Developmental foundations: Spelke's core knowledge theory [31, 32] posits domain-specific representational systems with a cascaded developmental trajectory: object individuation through spatiotemporal continuity [31] is a prerequisite for goal attribution [38, 9], which is a prerequisite for belief modeling [37, 5, 6]. Leslie [17] formalized this as a metarepresentational requirement; Onishi and Baillargeon [22] and Baillargeon *et al*. [4] demonstrated implicit false-belief sensitivity in infancy; Wellman *et al*. [36] meta-analyzed 178 studies confirming robust developmental change. Apperly and Butterfill [2] proposed dual-systems distinguishing efficient inflexible belief tracking from flexible effortful mentalizing; Shu *et al*. [30] operationalized core psychological reasoning in AGENT. This literature establishes the cascade as a property of human development; it does not test whether VLMs are subject to a parallel dependency. Our protocol provides that test, and the dose-response signature places the bottleneck in the front-end of the cascade.

Model-based ToM and structured intermediates: Several approaches assume access to structured state representations as input. ToMNet [26] trains a meta-learner over agent trajectories represented as state-action sequences; BIP-ALM [14] combines Bayesian inverse planning with structured scene representations to achieve strong belief inference. These systems take structured intermediates as given and evaluate the reasoning that follows. Our protocol takes the inverse view: it holds the inferential pathway constant and varies the fidelity of the structured intermediate that visual encoders attempt to produce, isolating the cost of extracting that intermediate from pixels.

3 Benchmarking Protocol

We treat visual ToM as two pipelines in series: a perceptual front-end converting pixels into propositions, and an inferential back-end mapping them and a question to an answer. A single accuracy figure under visual input cannot tell us which is failing. Our protocol forces them apart by holding the inferential task constant while parametrically varying perceptual fidelity, from no scaffold to a complete ground-truth scene description. It applies identically to abstract 2D panels and naturalistic 3D game scenes.

3.1 The Five-Condition Dissociation

Each probe is evaluated under five conditions, sampling the visual-to-text fidelity continuum. A and C anchor the endpoints; the three B conditions sit between, each adding a textual scaffold of increasing fidelity to the image.

A **Visual only:** Multi-panel image (abstract) or sampled video frames (naturalistic) with the question; no scene description. Measures how well the model's own front-end extracts the scene state the question requires.

B_a **Image with automatic-parser scaffold:** Visual input plus a scaffold from a parser with no ground-truth access. Abstract: a hand-engineered pixel pipeline using flood-fill blob detection, color matching, and inter-panel differencing (§A.2). Naturalistic: a frontier VLM (GPT-4o; additionally Claude Sonnet 4 on MindPower), prompted to produce panel-by-panel entity narrations with no ToM

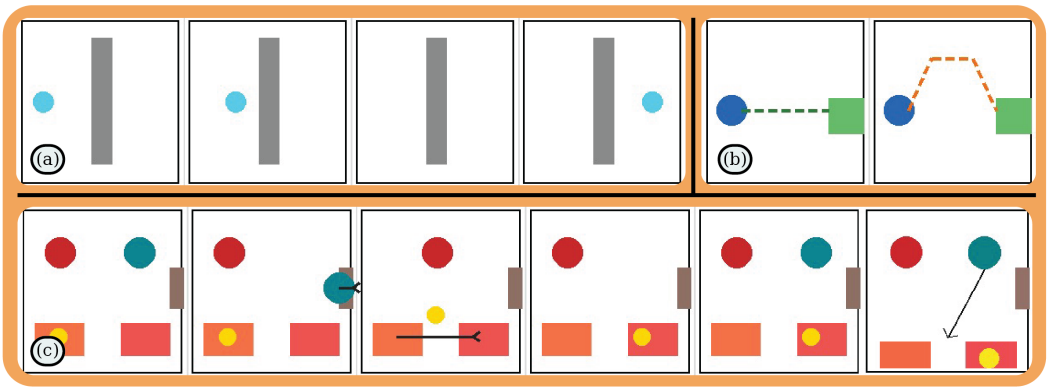


Figure 2: **Representative abstract 2D probes across the developmental cascade**, one template per stage. **(a)** Stage 1 object tracking (OCCLUSION_CONSISTENT): a cyan circle moves rightward, becomes hidden behind a gray occluder, and reappears. *Prompt:* "Is the final reappearance position consistent with the earlier motion direction?" (correct: yes). **(b)** Stage 2 goal attribution (WALL_REMOVED): a blue agent must reach a green goal with no obstacle present. *Prompt:* "Which path is more rational, the straight green path or the longer orange detour?" (correct: straight). **(c)** Stage 3 false belief (KNOWLEDGE_BELIEF): the teal character sees a yellow object in the left container and exits; the red character moves it to the right container; the teal character returns. *Prompt:* "Where will the teal character think the object is?" (correct: left). All stimuli rendered programmatically with no embedded text; full catalog in §H.

interpretation. Both are imperfect by construction; B_a asks whether a realistic front-end recovers the scene structure belief attribution requires.

B_p **Image with perceptual-grade scaffold:** Visual input plus a scaffold correct by construction but deliberately omitting visibility and absence inference: it names entities, positions, and observable events but never states that one agent did not observe what another did. This isolates how much scaffold benefit comes from perceptual content rather than visibility cues. Abstract: a panel-by-panel entity narration generated programmatically from the scene generator's state. MindPower: the dataset's human-written description, which contains observable scene state but not the belief, desire, intention, decision, or action layers used in scoring. Undefined for MuMA-ToM, whose annotations lack a comparable observable-only intermediate.

B_f **Image with full ground-truth scaffold:** Visual input plus the most complete reference scaffold per regime. Abstract: B_p plus explicit visibility and absence statements ("agent X was not in the scene when the object was moved"). MuMA-ToM: ground-truth entity identities, locations, actions, and visibility flags. MindPower: the full human reference chain across all six layers, shared across all scored layers of a scenario; on this regime, B_f functions as a full-information oracle upper bound rather than a leakage-controlled perceptual ceiling.

C **Text only:** The B_f scaffold with the image removed. Measures the reasoning ceiling when no visual processing is required; $\mathrm{Acc}(C) - \mathrm{Acc}(A)$ is the visual-to-text deficit that scaffolds attempt to close.

Successive scaffolds add content in interpretable steps: A $\rightarrow B_a$ adds whatever an automatic front-end produces; $B_a \rightarrow B_p$ makes the content correct but withholds visibility; $B_p \rightarrow B_f$ adds visibility and absence; $B_f \rightarrow C$ removes the image. Accuracy across the five conditions traces a dose-response curve along one axis: the fidelity of perceptual input reaching the inferential pathway.

3.2 Stimulus Regimes

The two regimes differ on every perceptual dimension. **Abstract 2D panels** present ToM structure in stripped-down form: multi-panel images of geometric shapes in the Heider-Simmel tradition [11], with zero embedded text. **Naturalistic 3D game scenes** move to multi-agent household activity rendered in 3D game environments, drawing on MuMA-ToM [29] and MindPower [39]. The two stress-test the perceptual-bottleneck account from opposite directions: abstract asks whether the



Figure 3: **Representative frames from the naturalistic multi-agent benchmarks.** *Top:* MuMA-ToM [29] household videos with human-rendered agents; questions probe BELIEF, BELIEF_OF_GOAL, SOCIAL_GOAL. **(a)** kitchen with red carpet and visible actor. **(b)** kitchen-adjacent layout with secondary spatial structure. **(c)** bedroom-to-living transition with distinct room geometry. **(d)** dimly lit living room with sofa and table layout. *Bottom:* MindPower [39] ThreeDWorld scenarios; the model acts as a robot helper inferring the human's implicit goal. **(e)** obstacle-removal: the owner enters the bedroom, picks up two of many pillows, and walks around. **(f)** missing-tool: the owner picks up bread at the dining table and walks back and forth as if missing something. **(g)** obstacle-removal: a wheelchair-using owner is blocked by a fire hydrant. **(h)** missing-tool: the owner holds a low-battery MacBook and cannot find the charging cable. Evaluations use multi-frame inputs (8 frames per MuMA-ToM video, 12-frame storyboards per MindPower scenario).

bottleneck appears even when stimuli are deliberately easy to parse; naturalistic asks whether the same dose-response reproduces under realistic clutter, occlusion, and multi-frame dynamics.

Abstract 2D panels: Following the developmental cascade in human cognitive development [31, 6, 36], the regime spans three stages, each requiring the previous as computational input (full diagram in §C). Stage 1 (Object Tracking, 6 templates, 300 probes) tests spatiotemporal individuation: occlusion continuity, trajectory-violation detection, count maintenance, permanence, color-change [31]. Stage 2 (Goal Attribution, 4 templates, 200 probes) tests inference of goals from motion, corresponding to 6–12 month infant capacities [38, 9]: target inference, detour reasoning, non-goal discrimination, efficient action prediction. Stage 3 (False Belief, 10 templates, 500 probes) tests the canonical ToM capacity [37, 5, 36]: Sally-Anne search and belief reports, reality-belief dissociation, knowledge access, unexpected contents, two-room transfers, perspective divergence, and second-order belief. Each template gets 50 randomized variants for 1,000 probes total. We follow construction conventions of prior work and do not claim the panel set as a contribution; representative probes in Figure 2; full template descriptions in §C.

Naturalistic 3D game scenes: MuMA-ToM [29] provides 225 sampled household-activity videos with multiple human-rendered agents performing belief-relevant actions, yielding 900 multiple-choice questions across three types: BELIEF (202), BELIEF_OF_GOAL (496), and SOCIAL_GOAL (202). The types are imbalanced; per-model overall accuracy is question-weighted across the 900 items, so it differs from the unweighted mean of per-type aggregates. **MindPower** [39] provides 492 photorealistic embodied scenarios (36 ThreeDWorld, 456 VirtualHome) with gold annotations across six layers: perception, belief, desire, intention, decision, action. Each layer is scored as the average of BERTScore-F1 and sentence-transformer cosine similarity against the gold reference. Frames are sampled at eight per video as a flattened multi-image input. Both datasets increase perceptual difficulty along every realistic axis (clutter, occlusion, multi-agent dynamics, multi-frame integration, lighting), inverting the abstract regime. Representative frames in Figure 3.

3.3 The Perceptual Contribution Ratio

The five conditions yield five accuracy numbers per model and benchmark. We collapse them to a single ratio: of the visual-to-text deficit $\mathrm{Acc}(C) - \mathrm{Acc}(A)$, what fraction does a scaffold of given fidelity recover?

Table 1: **Aggregate accuracy (%) by regime and condition.** 7-model means. (a) abstract 2D, by stage. (b) MuMA-ToM, by question type (overall is question-weighted; §3.2). (c) MindPower, layer-mean weighted across 36 TDW + 456 VirtualHome; B_a averages two parser variants (Table 5). **Bold**: best per row.

Stage / Type / Layer	A	B_a	B_p	B_f	C
(a) Abstract 2D panels: developmental stages					
1: Object Tracking	72.8	71.5	**96.4**	93.4	89.4
2: Goal Attribution	90.9	89.4	**99.2**	98.9	93.1
3: False Belief	73.9	75.7	88.8	**98.4**	94.5
(b) Naturalistic MuMA-ToM: by question type					
BELIEF	58.8	64.9	–	**74.7**	67.7
BELIEF_OF_GOAL	41.6	44.4	–	**52.9**	50.6
SOCIAL_GOAL	37.6	37.0	–	**53.6**	48.8
(c) Naturalistic MindPower: by reasoning layer					
PERCEPTION	19.3	21.6	26.8	24.8	**47.4**
BELIEF	23.4	22.7	28.8	35.5	**35.9**
DESIRE	21.4	20.0	24.9	**29.0**	28.7
INTENTION	21.4	19.2	25.6	28.7	**31.3**
DECISION	23.3	21.2	28.1	31.3	**36.2**
ACTION	6.1	4.9	10.2	14.4	**18.7**

Table 2: **PCR by capability tier across stimulus regimes** (95% bootstrap CIs in brackets). Abstract: Stage 3 false belief. MuMA-ToM: question-weighted aggregate. MindPower: scenario-weighted layer-mean across both ThreeDWorld and VirtualHome. **Bold**: best within each regime block.

Tier	$\mathrm{PCR}_a\uparrow$	$\mathrm{PCR}_p\uparrow$	$\mathrm{PCR}_f\uparrow$
(a) Abstract 2D panels, Stage 3			
Frontier	0.02 [-.12, .15]	0.96 [.89, 1.04]	**1.22** [1.14, 1.31]
Mid-scale	-1.10 [-1.45, -.81]	0.27 [.12, .40]	**1.01** [.99, 1.04]
Small-scale	0.82 [.73, .92]	0.77 [.67, .87]	**1.26** [1.18, 1.34]
(b) Naturalistic MuMA-ToM			
Frontier	0.25 [.10, .41]	–	**1.81** [1.61, 2.04]
Mid-scale	0.04 [-.18, .27]	–	**1.57** [1.32, 1.86]
Small-scale	0.53 [.38, .69]	–	**0.66** [.50, .83]
(c) Naturalistic MindPower			
Frontier	-0.06 [-.08, -.05]	0.39 [.37, .42]	**0.49** [.45, .53]
Mid-scale	-0.12 [-.15, -.09]	0.42 [.38, .46]	**0.86** [.80, .93]
Small-scale	-0.02 [-.04, .00]	0.23 [.20, .26]	**0.51** [.47, .56]

A VLM processes a stimulus I through perceptual extraction π, producing internal propositions, followed by reasoning ρ mapping propositions and a question q to an answer: $f_{\text{vis}}(I,q)=\rho(\pi(I),q)$. In text-only mode, $f_{\text{txt}}(T,q) = \rho(\tau(T),q)$, where τ encodes a narration T into the same propositional space. A scaffold s_x at fidelity x augments visual input via token-level fusion: $f_{\text{scaffold}}(I,s_x,q)=\rho(\pi(I)\oplus\tau(s_x),q)$. Treating ρ as shared across modalities is a simplification; the protocol tests its functional consequence rather than its mechanistic identity.

Definition 1 (Perceptual Contribution Ratio)**.**

$$\mathrm{PCR}_x = \frac{\mathrm{Acc}(B_x)-\mathrm{Acc}(A)}{\mathrm{Acc}(C)-\mathrm{Acc}(A)}, \tag{1}$$

where A is visual-only, B_x is image-with-scaffold at fidelity $x\in\{a,p,f\}$, and C is text-only.

The triplet $(\mathrm{PCR}_a,\mathrm{PCR}_p,\mathrm{PCR}_f)$ traces a dose-response curve. $\mathrm{PCR}_x=1$ indicates the scaffold closes the gap entirely (purely perceptual deficit); $\mathrm{PCR}_x=0$ indicates no benefit (purely conceptual deficit); intermediate values give the fraction of the deficit perceptual repair recovers; negative values flag misleading scaffolds; values above 1 indicate the scaffold-plus-image combination matches or exceeds the text-only ceiling (§4). The metric is undefined when $\mathrm{Acc}(C)=\mathrm{Acc}(A)$; the gaps observed throughout this paper make that case moot.

This functional form is not arbitrary. PCR is the unique metric on triples $(\mathrm{Acc}(A),\mathrm{Acc}(B_x),\mathrm{Acc}(C))$ with $\mathrm{Acc}(C)>\mathrm{Acc}(A)$ that satisfies endpoint calibration, invariance to positive affine reparameterizations, and affine dependence on the scaffolded accuracy (proof in §D.4). Any alternative form must abandon at least one assumption.

3.4 Models and Evaluation

Model lineup: We evaluate seven VLMs across three tiers reflecting parameter count and provider positioning. *Frontier* (top-line commercial, >100B): GPT-4o (Nov. 2024) [23], Gemini 2.5 Pro Preview [10], Claude Sonnet 4 [1]. *Mid-scale* (17B–72B): Qwen2.5-VL-72B-Instruct [3], Llama 4 Maverick [20]. *Small-scale* (sub-10B or mini): GPT-4.1-mini [24], Ministral-8B [21]. All seven are evaluated on both regimes. Models are accessed through the OpenRouter API with deterministic decoding (temperature 0, max 200 output tokens, single-turn); request parameters in §A.3.

Table 3: **Per-model accuracy (%) on abstract 2D Stage 3 (False Belief).** n=500 probes per model. Models are separated by tier (see §3.4). **Bold**: best per row.

Model	A	B_a	B_p	B_f	C
GPT-4o [23]	76.4	86.6	90.2	**99.2**	91.6
Gemini 2.5 Pro [10]	71.8	68.4	90.4	92.6	**95.4**
Claude Sonnet 4 [1]	78.8	73.2	97.6	**100.0**	93.2
Qwen2.5-VL-72B [3]	79.2	57.0	90.0	**100.0**	99.4
Llama 4 Maverick [20]	87.0	72.8	85.0	99.8	**100.0**
GPT-4.1-mini [24]	70.2	80.8	80.2	**100.0**	94.8
Ministral-8B [21]	53.6	91.2	88.4	**97.4**	87.0
Mean	73.9	75.7	88.8	**98.4**	94.5

Table 4: **Per-model accuracy (%) on MuMA-ToM** (n=900 probes/model; B_p undefined). **Bold**: best per row.

Model	A	B_a	B_f	C
GPT-4o [23]	45.8	47.6	52.4	**56.4**
Gemini 2.5 Pro [10]	62.3	59.1	**84.2**	65.1
Claude Sonnet 4 [1]	33.8	43.3	**64.9**	52.8
Qwen2.5-VL-72B [3]	48.1	45.1	**58.1**	55.8
Llama 4 Maverick [20]	44.2	47.8	**55.7**	50.2
GPT-4.1-mini [24]	41.1	46.9	51.4	**55.8**
Ministral-8B [21]	37.4	41.9	39.9	**42.1**
Mean	44.7	47.4	**58.1**	54.0

Answer grading: Responses are normalized (lowercased, whitespace-collapsed) before deterministic grading with no partial credit. For yes/no, the response must begin with the expected token; for numerical, the expected digit must appear in the first four characters; for directional, the response must contain any expected alias, including container color names, where unambiguous; for the reality-versus-belief template, the response is split at "belief" with the reality answer before and belief answer after. Identical across all conditions and models; alias lists and split-answer algorithms in §A.4. MuMA-ToM uses the multiple-choice grading from [29]; MindPower uses the per-layer scoring from [39].

Statistical methodology: Confidence intervals are computed by probe-level bootstrap with 10,000 iterations, resampling probe indices with replacement and reporting the 2.5th and 97.5th percentiles. PCR CIs use the same bootstrap, recomputing the ratio per iteration. Pairwise condition comparisons use McNemar's test on matched (model, probe) pairs with Bonferroni correction across the 12 protocol comparisons ($\alpha_{\text{Bonf}} = 0.0042$); for abstract Stage 3, this yields $n = 7 \times 500 = 3{,}500$ paired observations per comparison. Full detail in §D.

Scale: The protocol totals roughly 147,000 API calls: ∼99,000 on the abstract regime (including ablations on B_{vlm}, $B_{\text{vis}-\text{stripped}}$, chain-of-thought, and self-scaffold reported in §G) and ∼48,000 on the naturalistic regime. Parser-failure responses are excluded from totals.

4 Results

The protocol described in §3 was applied across the seven-model lineup on both stimulus regimes, totaling roughly 147,000 API calls. We organize the findings around four claims that the dose-response signature licenses: ground-truth scaffolds substantially close the visual-to-text gap on every regime; automatic parsers fail across regimes and tiers; the dose-response is monotonic in scaffold fidelity; and the dominant visual-only error mode is reality bias. Headline aggregate numbers appear in Table 1; per-tier PCR analysis in Table 2; per-model breakdowns in Tables 3, 4, and 5.

Ground-truth scaffolds substantially close the visual-to-text gap on every regime: On abstract 2D Stage 3, aggregate visual-only accuracy is 73.9% and text-only is 94.5%, a gap of 20.6 pp. The full ground-truth scaffold B_f closes this gap entirely, reaching 98.4%, with $\text{PCR}_f = 1.19$ (95% CI [1.09, 1.30]). The B_f-plus-image condition matches or exceeds the text-only ceiling for every tier (Table 2a): frontier 1.22, mid-scale 1.01, small-scale 1.26. Naturalistic MuMA-ToM shows the same pattern at a larger magnitude: aggregate $\text{PCR}_f = 1.44$, frontier 1.81, mid-scale 1.57 (Table 2b). The image-plus-scaffold outperforming text-only is consistent with the scaffold restoring the perceptual content the front-end failed to extract, while the visual input continues to contribute non-redundant information; we do not commit to a stronger interpretation of $\text{PCR} > 1$ here. On MindPower, the picture changes quantitatively but not qualitatively: B_f recovers 59% of a 13.9 pp visual-to-text gap, with mid-scale strongest at $\text{PCR}_f = 0.86$. On this regime, B_f is an oracle full-information scaffold (§3.1); the leakage-controlled B_p recovers 39% of the gap on the frontier and 42% on mid-scale (Table 2c). Across all three regimes, every PCR_f is positive and statistically distinct from zero ($p_{\text{Bonf}} < 10^{-4}$, McNemar A vs B_f).

Table 5: **Per-model layer-mean accuracy (%) on MindPower.** (a) ThreeDWorld (36 scenarios). (b) VirtualHome (456 scenarios). B_a^G and B_a^C: GPT-4o and Claude Sonnet 4 as automatic parsers. B_f is an oracle full-information scaffold containing the human reference chain across all six layers (§3.1). Tier abbreviations: F = Frontier, M = Mid-scale, S = Small-scale. **Bold**: best per row.

Model	**Tier**	A	B_a^G	B_a^C	B_p	B_f	C
(a) ThreeDWorld (36 scenarios)							
GPT-4o [23]	F	31.6	30.7	26.9	45.1	54.9	**54.9**
Gemini 2.5 Pro [10]	F	35.7	34.6	33.5	41.9	46.4	**57.1**
Claude Sonnet 4 [1]	F	25.5	31.3	26.3	45.4	41.6	**55.8**
Qwen2.5-VL-72B [3]	M	28.8	29.2	26.1	49.0	44.5	**49.1**
Llama 4 Maverick [20]	M	24.7	28.8	25.3	44.5	**51.9**	51.2
GPT-4.1-mini [24]	S	32.6	31.5	29.9	45.2	45.1	**56.6**
Ministral-8B [21]	S	20.2	23.3	20.4	27.7	35.9	**38.2**
Mean		28.4	29.9	26.9	42.7	45.8	**51.9**
(b) VirtualHome (456 scenarios)							
GPT-4o [23]	F	19.3	17.1	16.9	23.5	**31.7**	29.5
Gemini 2.5 Pro [10]	F	19.4	18.4	18.3	24.5	22.8	**34.2**
Claude Sonnet 4 [1]	F	18.3	18.5	18.9	25.8	23.0	**37.5**
Qwen2.5-VL-72B [3]	M	19.4	18.2	17.3	23.4	26.8	**31.1**
Llama 4 Maverick [20]	M	18.4	17.2	16.7	21.7	**29.3**	28.1
GPT-4.1-mini [24]	S	19.8	18.8	18.8	23.4	25.5	**35.3**
Ministral-8B [21]	S	14.4	14.6	14.9	15.9	21.6	**25.2**
Mean		18.4	17.5	17.4	22.6	25.8	**31.5**

Automatic parsers fail across regimes: Where ground-truth scaffolds rescue performance, automatic-parser scaffolds do not (Table 2, PCR_a column). On abstract 2D Stage 3, the aggregate PCR_a is 0.09, with the mid-scale tier at -1.10: the parser actively misleads. The hand-engineered abstract parser correctly recovers entity identities and positions on 99.4% of probes (§B.2), so the failure is not at low-level extraction. The deficit is that parser output, even when correct on entity-level facts, lacks the visibility and presence-absence structure false-belief requires. On MindPower we ran two independent parsers: GPT-4o-as-parser yields aggregate $\mathrm{PCR}_a = -0.06$ across tiers and Claude-as-parser yields -0.10 (Table 5). Failure consistency across two frontier parsers, in addition to the hand-engineered parser on abstract panels, is the diagnostic: the bottleneck is not bridged by current visual encoders in zero-shot scene-describer roles.

The dose-response is monotonic in scaffold fidelity: Where all three scaffold conditions are defined (abstract 2D Stage 3 and MindPower), accuracy increases monotonically from A through B_a through B_p through B_f, the only exception being B_a which falls slightly below A on both regimes for the reasons above. Abstract Stage 3: $A = 73.9 \rightarrow B_a = 75.7 \rightarrow B_p = 88.8 \rightarrow B_f = 98.4$. MindPower (overall layer-mean): $A = 19.1 \rightarrow B_a = 18.3 \rightarrow B_p = 24.1 \rightarrow B_f = 27.3$. The ordering rules out non-perceptual explanations: if scaffolds helped through token-count or format-priming, B_a would close more of the gap and B_p would not be reliably below B_f. Monotonicity holds at the per-model level for 5 of 7 models on abstract Stage 3 (Table 3), with deviations driven by ceiling effects on Claude Sonnet 4 and Llama 4 Maverick. We additionally ran a stripped variant $B_{\mathrm{vis-stripped}}$ on abstract, removing all explicit visibility and absence statements from B_f via regex (§G.2; 8.9% of scaffold characters removed). Aggregate accuracy is 90.0%, $\mathrm{PCR}_{\mathrm{vis-stripped}} = 0.79$, recovering most of the B_f benefit. Structured perceptual content alone, without the visibility statements that name the false-belief inference, recovers a substantial fraction of the gap.

The dominant visual-only error mode is reality bias: We categorized all 915 visual-only errors on abstract Stage 3 (7 models $\times$ 500 probes) using a deterministic rule (§A.5). An error is a *reality-bias* error if the model reports the object's veridical current location rather than the location consistent with the absent agent's belief. Reality-bias errors account for 549 of 915 visual-only errors (60.0%); the remaining 40% are uncategorized. The reality-bias share is invariant across tiers within 5 pp. The bias is not a failure of belief reasoning in the abstract: under the same probes, text-only accuracy is 94.5%. It appears only when the question is conveyed visually and disappears when the

visibility-and-absence content is provided in the scaffold. The mechanism is a perceptual front-end that encodes *what* an object is and *where* it currently is but not *who was present when it moved*, leaving the inferential pathway with insufficient state to dissociate reality from belief. Robustness checks on prompt phrasing, chain-of-thought self-prompting, and self-generated scaffolds confirm the dose-response is not an artifact of any specific prompt, decoding strategy, or grading rule (§G).

5 Discussion

The bottleneck is perceptual, not conceptual: The four findings converge on a single picture. When the scene is given in pixels, frontier VLMs fail false-belief tasks by margins commonly attributed to a missing belief-reasoning capacity. When the same problem is given in text, the same models succeed. When the scene is given in pixels alongside the structured perceptual content the visual front-end should have produced but did not, performance recovers to within the text-only ceiling on every regime. The deficit is not in the inferential pathway that operates on scene representations: that pathway is intact, as evidenced by the text-only ceiling. The deficit is in the front-end that converts pixels to scene representations. Reality bias names the specific failure: the front-end successfully encodes object identity and current location but does not encode who was present at each moment, leaving the inferential pathway with the wrong propositional input. Asking the model to reason harder about belief is asking the wrong system to do work that the perceptual system has not finished.

Reconciliation with prior accounts: Several recent works have argued for a cognitive or representational diagnosis of visual ToM failures in VLMs [19, 8, 34, 13]. We do not contest these accounts on their original tasks. Liu *et al*. [19] demonstrate attention-vs-output dissociations on single-image VQA where the visual evidence is spatially localized in one frame; Chen *et al*. [8] document below-chance perspective-taking on static scenes; Tong *et al*. [34] trace failures to CLIP-based representations. These dissociations can coexist with our claim because the tasks differ in perceptual extraction complexity. Our protocol distinguishes the two diagnoses on a per-task basis: dose-response with PCR_p or PCR_f near 1 localizes the deficit to perceptual extraction; a flat or near-zero dose-response would localize it to inference. For sequential multi-panel and multi-frame ToM tasks, the signature consistently localizes to perception. The result is also consistent with model-based ToM approaches that take structured scene representations as given [26, 14]: those methods bypass the front-end by construction; our protocol provides the missing measurement of how much the front-end costs to skip.

Architectural implications: If the bottleneck is in front-end scene-state extraction rather than belief inference, the prescription is not new reasoning modules but better perceptual representations. The specific competences current encoders fail to produce are object-level entity tracking with persistent identity across frames, and presence-absence registers indexed to entities at each frame. Both are familiar from object-permanence and dynamic-occluder reasoning in cognitive development [31, 4] and from object-centric representation learning. We predict that VLM architectures with explicit object slots, cross-frame correspondence, and visibility tracking would close most of the visual-to-text gap on existing ToM benchmarks without requiring belief-specific training objectives. Pretraining objectives that incentivize per-entity state extraction, multi-agent occlusion tracking, and event-level presence accounting, rather than global scene captioning, are a concrete path forward.

6 Conclusion

We proposed a five-condition dissociation protocol that holds the inferential task constant while varying perceptual fidelity, instantiated on abstract 2D panels and naturalistic 3D game scenes, and quantified the result with the Perceptual Contribution Ratio. Across seven VLMs and roughly $147{,}000$ API calls, the visual ToM deficit in frontier models is predominantly perceptual: ground-truth scaffolds close the visual-to-text gap on every regime, automatic parsers including frontier VLMs as zero-shot scene describers fail, and the dominant visual-only error is reality bias, the signature of a front-end that fails to encode presence and absence across frames. The route to better visual social reasoning runs through perceptual representations that track entities and who is present when, not belief-specific reasoning modules layered on current encoders.

Acknowledgements

Juergen Gall has been supported by the Federal Ministry of Research, Technology and Space (BMFTR) under grant no. 16IS22094A WEST-AI and the ERC Consolidator Grant FORHUE (101044724).

References

[1] Anthropic. Claude Sonnet 4, 2025. `https://www.anthropic.com/claude`.

[2] Ian A Apperly and Stephen A Butterfill. Do humans have two systems to track beliefs and belief-like states? *Psychological Review*, 116(4):953–970, 2009.

[3] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, Junyang Lin, et al. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025.

[4] Renée Baillargeon, Rose M Scott, and Zijing He. False-belief understanding in infants. *Trends in Cognitive Sciences*, 14(3):110–118, 2010.

[5] Simon Baron-Cohen, Alan M Leslie, and Uta Frith. Does the autistic child have a "theory of mind"? *Cognition*, 21(1):37–46, 1985.

[6] Susan Carey. *The Origin of Concepts*. Oxford University Press, 2009.

[7] Kanishk Gandhi, Jan-Philipp Fränken, Tobias Gerstenberg, and Noah D Goodman. Understanding social reasoning in language models with language models. In *Advances in Neural Information Processing Systems*, volume 37, 2023.

[8] Qingying Gao, Yijiang Li, Haiyun Lyu, Haoran Sun, Dezhi Luo, and Hokin Deng. Vision language models see what you want but not what you see. *arXiv preprint arXiv:2410.00324*, 2024. ICLR 2025 Workshop on Bidirectional Human-AI Alignment (BiAlign).

[9] György Gergely, Zoltán Nádasdy, Gergely Csibra, and Szilvia Bíró. Taking the intentional stance at 12 months of age. *Cognition*, 56(2):165–193, 1995.

[10] Google DeepMind. Gemini 2.5 Pro, 2025. `https://deepmind.google/technologies/gemini/`.

[11] Fritz Heider and Marianne Simmel. An experimental study of apparent behavior. *The American Journal of Psychology*, 57(2):243–259, 1944.

[12] Wenyi Hong, Yean Cheng, Zhuoyi Yang, Weihan Wang, Lefan Wang, Xiaotao Gu, Shiyu Huang, Yuxiao Dong, and Jie Tang. Motionbench: Benchmarking and improving fine-grained video motion understanding for vision language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8450–8460, 2025.

[13] Jennifer Hu, Felipe Sosa, and Tomer Ullman. Re-evaluating theory of mind evaluation in large language models. *Philosophical Transactions of the Royal Society B*, 2025.

[14] Chuanyang Jin, Yutong Wu, Jing Cao, Jiannan Xiang, Yen-Ling Kuo, Zhiting Hu, Tomer Ullman, Antonio Torralba, Joshua B Tenenbaum, and Tianmin Shu. MMToM-QA: Multimodal theory of mind question answering. *arXiv preprint arXiv:2401.08743*, 2024.

[15] Hyunwoo Kim, Melanie Sclar, Xuhui Zhou, Ronan Le Bras, Gunhee Kim, Yejin Choi, and Maarten Sap. FANToM: A benchmark for stress-testing machine theory of mind in interactions. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 14397–14413, 2023.

[16] Michal Kosinski. Evaluating large language models in theory of mind tasks. *Proceedings of the National Academy of Sciences*, 121(45), 2024.

[17] Alan M Leslie. Pretense and representation: The origins of "theory of mind". *Psychological Review*, 94(4):412–426, 1987.

[18] Yijiang Li, Qingying Gao, Haoran Sun, Haiyun Lyu, Dezhi Luo, and Hokin Deng. Cogdevelop2k: Reversed cognitive development in multimodal large language models. *arXiv preprint arXiv:2410.10855*, 2024.

[19] Zhining Liu, Ziyi Chen, Hui Liu, Chen Luo, Xianfeng Tang, Suhang Wang, Jingying Zeng, Zhenwei Dai, Zhan Shi, Tianxin Wei, Hanqing Lu, Benoit Dumoulin, and Hanghang Tong. Seeing but not believing: Probing the disconnect between visual attention and answer correctness in VLMs. In *International Conference on Learning Representations (ICLR)*, 2026.

[20] Meta AI. Llama 4, 2025. `https://ai.meta.com/llama/`.

[21] Mistral AI. Ministral 8B: A small frontier model from Mistral AI, 2024. `https://mistral.ai/news/ministraux/`.

[22] Kristine H Onishi and Renée Baillargeon. Do 15-month-old infants understand false beliefs? *Science*, 308(5719):255–258, 2005.

[23] OpenAI. GPT-4o, 2024. `https://openai.com/gpt-4o`.

[24] OpenAI. GPT-4.1 mini, 2025. `https://openai.com/index/gpt-4-1/`.

[25] David Premack and Guy Woodruff. Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences*, 1(4):515–526, 1978.

[26] Neil Rabinowitz, Frank Perbet, Francis Song, Chiyuan Zhang, S. M. Ali Eslami, and Matthew Botvinick. Machine theory of mind. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 4218–4227, 2018.

[27] Maarten Sap, Ronan LeBras, Daniel Fried, and Yejin Choi. Neural theory-of-mind? On the limits of social intelligence in large LMs. *arXiv preprint arXiv:2210.13312*, 2022.

[28] Natalie Shapira, Mosh Levy, Seyed Hossein Alavi, Xuhui Zhou, Yejin Choi, Yoav Goldberg, Maarten Sap, and Vered Shwartz. Clever hans or neural theory of mind? stress testing social reasoning in large language models. *arXiv preprint arXiv:2305.14763*, 2024.

[29] Haojun Shi, Suyu Ye, Xinyu Fang, Chuanyang Jin, Leyla Isik, Yen-Ling Kuo, and Tianmin Shu. MuMA-ToM: Multi-modal multi-agent theory of mind. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025.

[30] Tianmin Shu, Abhishek Bhandwaldar, Chuang Gan, Kevin A Smith, Shari Liu, Dan Gutfreund, Elizabeth Spelke, Joshua B Tenenbaum, and Tomer D Ullman. AGENT: A benchmark for core psychological reasoning. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pages 9614–9625, 2021.

[31] Elizabeth S Spelke. Principles of object perception. *Cognitive Science*, 14(1):29–56, 1990.

[32] Elizabeth S Spelke. Core knowledge. *American Psychologist*, 55(11):1233–1243, 2000.

[33] James W. A. Strachan, Dalila Albergo, Giulia Borghini, Oriana Pansardi, Eugenio Scaliti, Saurabh Gupta, Krati Saxena, Alessandro Rufo, Stefano Panzeri, Guido Manzi, Michael S. A. Graziano, and Cristina Becchio. Testing theory of mind in large language models and humans. *Nature Human Behaviour*, 8:1285–1295, 2024.

[34] Shengbang Tong, Zhuang Liu, Yuexiang Zhai, Yi Ma, Yann LeCun, and Saining Xie. Eyes wide shut? exploring the visual shortcomings of multimodal LLMs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9568–9578, 2024.

[35] Tomer Ullman. Large language models fail on trivial alterations to theory-of-mind tasks. *arXiv preprint arXiv:2302.08399*, 2023.

[36] Henry M Wellman, David Cross, and Julanne Watson. Meta-analysis of theory-of-mind development: The truth about false belief. *Child Development*, 72(3):655–684, 2001.

[37] Heinz Wimmer and Josef Perner. Beliefs about beliefs: Representation and constraining function of wrong beliefs in young children's understanding of deception. *Cognition*, 13(1):103–128, 1983.

[38] Amanda L Woodward. Infants selectively encode the goal object of an actor's reach. *Cognition*, 69(1):1–34, 1998.

[39] Ruoxuan Zhang, Qiyun Zheng, Zhiyu Zhou, Ziqi Liao, Siyu Wu, Jian-Yu Jiang-Lin, Bin Wen, Hongxia Xie, Jianlong Fu, and Wen-Huang Cheng. Mindpower: Enabling theory-of-mind reasoning in VLM-based embodied agents. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

Supplementary Material

Perception, Not Reasoning, Limits Visual Theory of Mind

A B_a B_p B_f C

This supplementary is organized around a single evidentiary chain: visual false-belief failures persist when models must extract state from pixels, but largely disappear when the missing perceptual state is supplied as text. Section A fixes the protocol and accounting. Section B shows the full stage and template breakdowns. Section C documents the complete task family. Section D explains the matched uncertainty estimates and gives the axiomatic characterization of PCR. Section E relates the claim to competing accounts. Section F states the remaining caveats. Section G closes alternative explanations with targeted ablations, including VLM-as-parser, visibility stripping, paraphrases, chain-of-thought, and self-scaffold baselines.

A Implementation Details

A.1 Stimulus Generation

All 1,000 stimuli are generated programmatically using PIL (Python Imaging Library). Each panel measures 200×200 pixels. Multi-panel stimuli are created by concatenating panels horizontally, yielding images of width $200 \times N_{\text{panels}}$ where N_{panels} is the panel count (4–6 depending on template). Adjacent panels are separated by a 2-pixel light gray (`#DDDDDD`) divider.

Entity rendering parameters. Characters (agents) are rendered as filled circles with radius 18 px. Objects are smaller filled circles with radius 10 px (12 px for Stage 1 templates). Containers are filled rectangles of 58×38 px. Obstacles and doors are gray (`#888888`) or brown (`#8D6E63`) rectangles. Room boundaries are drawn as 2-pixel black outlines.

Color palettes. Six agent colors: blue (`#1565C0`), red (`#C62828`), green (`#2E7D32`), purple (`#6A1B9A`), orange (`#E65100`), teal (`#00838F`). Five object colors: yellow (`#FFD600`), amber (`#FF6F00`), lime (`#76FF03`), pink (`#E040FB`), cyan (`#00E5FF`). Six container colors: light-blue (`#42A5F5`), coral (`#EF5350`), brown (`#5D4037`), mint (`#66BB6A`), violet (`#AB47BC`), peach (`#FF7043`). For each probe variant, character colors, object colors, and container colors are sampled without replacement from their respective palettes using a seeded random number generator (seed = $10{,}000 \times (\text{template_index} + 1) + \text{variant_index}$), ensuring full reproducibility.

Zero text constraint. No text labels, panel numbers, character names, or narration appear in any stimulus image.

A.2 Automatic Vision Parser

The parser operates on the saved PNG image with no access to generation metadata.

Foreground masking. A pixel is classified as foreground if it is not white (RGB > 210 on all channels), not black (< 50 on all channels), not gray (channel spread < 30 and mean between 80–210), and not door-brown ($|\text{R} - 141| < 25 \wedge |\text{G} - 110| < 25 \wedge |\text{B} - 99| < 25$).

Connected component extraction. Flood-fill BFS from each unvisited foreground pixel, expanding to 4-connected neighbors within $\Delta_{\text{RGB}} < 60$ from the seed pixel. Components smaller than 40 pixels are discarded. Components are capped at 6,000 pixels to prevent runaway fills.

Entity classification. Each blob is classified by area and circularity (area / bounding-box area):

- **Agent**: circular ($0.55 \leq \text{circularity} \leq 0.90$, aspect ratio 0.65–1.35) and area 400–2,000 px.
- **Object**: circular and area 100–550 px.
- **Container**: non-circular and area $\geq 1{,}500$ px.

Color identity is assigned by nearest-neighbor matching against the known palette (Euclidean distance in RGB, threshold 80).

Event inference. Agent presence is compared across consecutive panels. If an agent present in panel k is absent in panel $k{+}1$, an exit event is inferred. Object position (left/right of panel center) is tracked to infer transfer events.

Scaffold serialization. Parsed structure is serialized as: "Panel k: agents: [name] agent ([side]-[vertical]). objects: [name] object ([side]-[vertical]). containers: [name] container ([side]-[vertical])."

A.3 API Call Protocol

All models are accessed via the OpenRouter API (`https://openrouter.ai/api/v1/chat/completions`).

Request parameters. Temperature: 0.0. Max tokens: 200. Single-turn conversation (one user message). Images are included as base64-encoded PNG in the `image_url` field for conditions A, B_a, B_p, B_f, and all controls. Condition C sends text only (no image).

Retry logic. HTTP 429 (rate limit): exponential backoff starting at 30 s. HTTP 5xx: retry after 15 s. Maximum 3 retries per call. Failed calls are recorded as incorrect with an `ERROR` response prefix.

Parallelization. Multiple API keys support parallel workers. Task queue distributes probe-model-condition triples. Results are checkpointed every 100 completions.

Call accounting and cost. The paper-facing benchmark used 108,500 API calls: 60,200 for the Abstract 2D panels experiment (five conditions, three adversarial controls, three prompt paraphrases) and 48,300 for the naturalistic 3D game-scene evaluations on MuMA-ToM and MindPower. The corresponding paper-facing cost was $319.83: $65.69 for the Abstract 2D panels component and $254.14 for MuMA-ToM and MindPower. With the 38,500 Abstract 2D panels ablation calls in §G, the full audit trail comprises $\sim$147,000 API calls and $369.83.

A.4 Answer Grading

Responses are normalized (lowercased, whitespace-collapsed) before grading. Grading is answer-type-specific:

- **yes_no**: Response must start with the expected token ("yes" or "no").
- **number**: Expected digit must appear in the first 4 characters.
- **left_right**: Response must contain any of the expected aliases (e.g., "left", "left box", container color name).
- **belief_split** (reality-vs-belief template): The response is split at the word "belief" (or at midpoint if absent); the reality answer must appear before the split and the belief answer after.
- **All other types**: Response must contain at least one expected alias.

No partial credit is awarded. The grading function is deterministic and identical across all conditions and models.

A.5 Error Categorization

Stage 3 Condition A errors are classified by a deterministic rule applied to the normalized model output. A response is labeled a **reality-bias** error if it contains an "actual alias" (the object's veridical location or the reality-consistent answer) but does not contain the expected belief-consistent answer; all remaining errors are uncategorized. Reality-bias errors account for 549 of 915 visual-only Stage 3 errors across the seven-model lineup (60.0%); the rule is identical across models.

A.6 Prompt Paraphrase Variants

Three paraphrases of each Stage 3 Condition A question are constructed:

- **Restructured**: "This image has [N_{panels}] panels showing colored figures, containers, and smaller objects. Follow the sequence carefully before answering: [task_prompt] [answer_instruction]"

Table S1: **Stages 1–2 per-model results** across the seven-model lineup (all five conditions, accuracy ↑ %). **Bold**: best per row; *italic*: 2nd.

	Stage 1: Object Tracking					Stage 2: Goal Attribution				
Model	A	B_a	B_p	B_f	C	A	B_a	B_p	B_f	C
GPT-4o [23]	83.3	83.3	**100**	**100**	83.3	93.0	91.5	**100**	**100**	**100**
Gemini 2.5 Pro [10]	71.0	83.3	83.7	80.3	**94.0**	*94.0*	**96.0**	*94.5*	92.5	**100**
Claude Sonnet 4 [1]	33.3	50.0	*99.3*	*99.3*	**100**	75.0	75.0	**100**	**100**	51.5†
Qwen2.5-VL-72B [3]	83.3	73.7	**100**	*94.7*	**100**	**100**	**100**	**100**	**100**	**100**
Llama 4 Mav. [20]	81.3	80.3	**100**	**100**	*98.0*	83.0	81.0	**100**	*99.5*	**100**
GPT-4.1-mini [24]	81.7	82.0	**100**	**100**	83.3	91.0	85.0	**100**	**100**	**100**
Ministral-8B [21]	75.7	47.7	**91.7**	*79.3*	67.0	**100**	97.5	**100**	**100**	**100**
Mean	72.8	71.5	**96.4**	*93.4*	89.4	90.9	89.4	**99.2**	*98.9*	93.1

† Claude text-only Stage 2 anomaly: direction-convention misinterpretation in pure-text probes (see §F).

Table S2: **Stage 3 per-template Condition A accuracy ↑ (%)** across the seven-model lineup (350 instances per template), sorted by difficulty. Parser: structural parse accuracy (%).

Template	Acc ↑	Parser ↑
unexpected_contents	55.4	100
two_rooms	56.0	94
second_order	64.9	100
reality_vs_belief	70.0	100
knowledge_belief	72.0	100
sally_anne_reality	74.9	100
knowledge_who_knows	80.6	100
sally_anne_search	83.4	100
sally_anne_belief	85.7	100
perspectives_differ	95.7	100
Mean	**73.9**	**99.4**

- **Vocabulary**: Synonym substitution applied to the original question ("character" → "figure", "container" → "box", "object" → "item", "search" → "look in").
- **Distractor**: "The background is white. [original question]"

B Extended Results

B.1 Stages 1–2 Per-Model Results

Table S1 presents the complete Stages 1–2 per-model matrix showing all five conditions.

B.2 Per-Template Stage 3 Results

Table S2 presents accuracy for all 10 Stage 3 templates.

B.3 Prompt Robustness Extended

The three paraphrase variants are applied to Stage 3 Condition A and compared against the 7-model abstract Stage 3 baseline (73.9%). All three variants produce accuracy within a narrow band of the original. Updated per-variant accuracies and signed deltas are reported alongside the paraphrase analysis for C and B_f in §G.3; the inferential competence captured in the matched conditions is robust to phrasing.

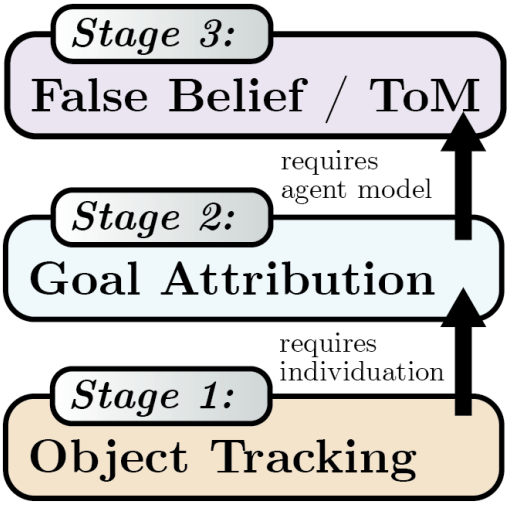


Figure S1: **Developmental stage hierarchy.** Each stage requires the representational output of the previous one, mirroring the cascade identified in human cognitive development: object individuation at 3–5 months [31], goal attribution at 6–12 months [38, 9], and false-belief reasoning at $\sim$4 years [37, 5]. 20 templates $\times$ 50 variants = 1,000 probes.

C Complete Template Catalog

The 20 templates are organized into three stages mirroring the developmental cascade in human cognitive development (Figure S1): object individuation must precede goal attribution, which must precede belief modeling. Each stage's templates are described below.

C.1 Stage 1: Object Tracking (6 templates)

occlusion_consistent / occlusion_violation. 5 panels. A colored circle moves rightward toward a gray barrier, disappears behind it (Panels 3–4), and reappears on the correct side (consistent) or the wrong side (violation). Question: "Is the final reappearance position consistent with the earlier direction of motion?" Answer: yes/no.

count_two / count_three. 4 panels. Two (or three) identically or distinctly colored circles follow crossing trajectories. Question: "How many distinct circular objects are present?" Answer: 2 or 3.

permanence. 4 panels. A colored circle enters a container (Panels 1–2), remains hidden (Panels 3–4). Question: "Is the object still present behind the occluder?" Answer: yes.

color_change. 5 panels. A colored circle moves behind a barrier and reappears in a different color. Question: "Did the object change color?" Answer: yes.

C.2 Stage 2: Goal Attribution (4 templates)

goal_upper. 4 panels. An agent moves along a trajectory toward one of two targets (upper or lower). Question: "Which target is the agent heading toward?" Answer: upper/lower.

detour_obstacle. 3 panels. An agent takes a curved path around a gray obstacle to reach a target. Question: "Why did the agent take a curved path?" Answer: obstacle.

no_goal. 4 panels. A circle drifts at constant velocity without changing direction toward any target. Question: "Is this motion goal-directed?" Answer: no.

***wall_removed*.** 2 panels. Left panel: straight path with no obstacle. Right panel: curved detour path with no obstacle. Question: "With no obstacle present, which path would a goal-directed agent choose?" Answer: straight.

C.3 Stage 3: False Belief / Theory of Mind (10 templates)

***sally_anne_search* / *sally_anne_belief* / *sally_anne_reality*.** 6 panels. Classic Sally-Anne scenario: Agent 1 places an object in Container L, exits, Agent 2 transfers the object to Container R, Agent 1 returns. Three question variants probe search prediction (where will Agent 1 look first? Left), belief report (where does Agent 1 think the object is? Left), and reality check (where is the object actually? Right).

***reality_vs_belief*.** 6 panels. Same scenario. The model must simultaneously report both the object's actual location (right) and Agent 1's believed location (left) in a single response.

***knowledge_who_knows* / *knowledge_belief*.** 5 panels. Two agents are in a room; one exits, the object is moved, the exiting agent returns. *knowledge_who_knows*: "Which character knows the object's current location?" (the one who stayed). *knowledge_belief*: "Where does the returning character think the object is?" (left, the pre-transfer location).

***unexpected_contents*.** 5 panels. Agent 1 sees a box with round symbols, opens it to find rectangular contents, closes it. Agent 2 arrives and sees only the closed box with round symbols. Question: "What does Agent 2 think is inside?" Answer: round objects.

***two_rooms*.** 5 panels. Two rooms divided by a wall. Agent 1 carries an object from the left room to the right room while Agent 2 watches. Agent 2 then exits. While Agent 2 is gone, Agent 1 places the object in the right container. Agent 2 returns. Question: "Where does Agent 2 think the object is?" Answer: right (Agent 2 saw the transfer to the right room before leaving).

***perspectives_differ*.** 5 panels. Similar to Sally-Anne but the question asks whether two agents hold the same or different beliefs about the object's location. Answer: different.

***second_order*.** 5 panels. One agent is absent during a transfer but observed by a third party. Question: "Does the third-party agent know that the absent agent holds a false belief?" Answer: yes.

D Statistical Methodology

D.1 Bootstrap Confidence Intervals

All confidence intervals are computed via probe-level bootstrap resampling. For each condition, the accuracy matrix has dimensions $M \times N_{\text{probes}}$ (models $\times$ probes). We resample N_{probes} probe indices with replacement 10,000 times, compute the mean accuracy across all models and resampled probes for each bootstrap iteration, and report the 2.5th and 97.5th percentiles as the 95% CI.

D.2 McNemar Tests

McNemar's test is applied to matched pairs: for each (model, probe_id) combination, we compare whether the response was correct under condition X vs. condition Y, yielding a 2×2 contingency table. The test statistic is $\chi^2 = (b-c)^2/(b+c)$, compared against χ^2_1. We apply Bonferroni correction across 12 pairwise comparisons ($\alpha_{\text{Bonf}} = 0.05/12 = 0.0042$). Stage 3 yields $n = 7 \times 500 = 3{,}500$ paired observations per comparison.

D.3 PCR Bootstrap

The Perceptual Contribution Ratio is bootstrapped by resampling probe indices and computing $\text{PCR}_x = (\overline{\text{Acc}}_B - \overline{\text{Acc}}_A)/(\overline{\text{Acc}}_C - \overline{\text{Acc}}_A)$ for each bootstrap sample.

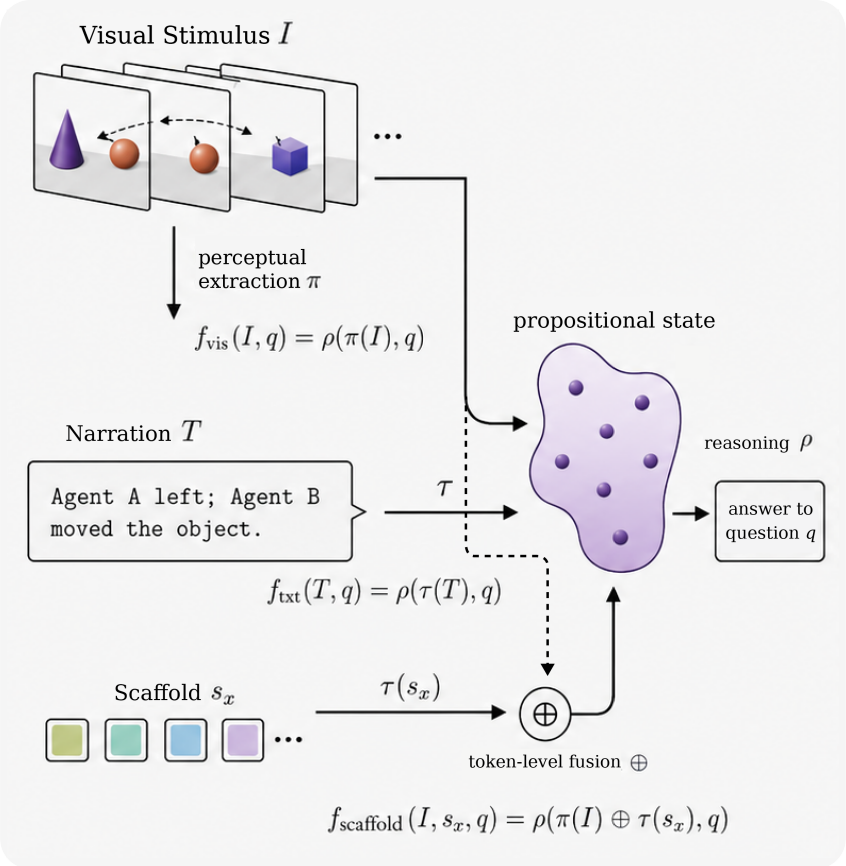


Figure S2: **Three pathways through a shared reasoning module ρ.** Visual input I enters via perceptual extraction π (f_{vis}); a narration T enters via the text encoder τ (f_{txt}); a scaffold s_x augments the visual path through token-level fusion $\oplus$ (f_{scaffold}). The five-condition protocol (§3) selects which pathway is active.

D.4 Axiomatic Characterization of PCR

The Perceptual Contribution Ratio (Definition 1) admits a uniqueness characterization from four natural axioms. The argument formalizes PCR as the *share of the closable visual-to-text gap recovered by the scaffold*, and shows that any metric respecting a small set of natural invariances must coincide with PCR.

Let M be a candidate metric defined on triples (a,b,c) with $c>a$, where $a=\text{Acc}(A)$, $b=\text{Acc}(B_x)$, and $c=\text{Acc}(C)$. The pathways formalized in Figure S2 (also stated in §3.3) supply the operational meaning: A corresponds to f_{vis}, C to f_{txt}, and B_x to f_{scaffold} at fidelity x. Consider the following axioms.

- (A1) **Endpoint calibration.** $M(a,a,c)=0$ and $M(a,c,c)=1$.
- (A2) **Translation invariance.** $M(a+\delta,b+\delta,c+\delta)=M(a,b,c)$ for all $\delta\in\mathbb{R}$.
- (A3) **Positive scale invariance.** $M(\alpha a,\alpha b,\alpha c)=M(a,b,c)$ for all $\alpha>0$.
- (A4) **Linearity in the scaffolded condition.** For fixed a,c, the map $b\mapsto M(a,b,c)$ is affine.

Together, axioms (A2) and (A3) are equivalent to invariance under positive affine maps $z\mapsto\alpha z+\delta$ with $\alpha>0$.

Interpretation. (A1) anchors the metric to the construct: a scaffold that fails to improve over visual-only contributes nothing (purely conceptual deficit), while a scaffold that reaches the text-only ceiling has fully closed the perceptual gap. (A2) and (A3) express that accuracy is an

interval-scale quantity with neither a canonical zero (chance levels differ across question formats and answer cardinalities) nor a fixed unit (proportions vs. percentages); the perceptual-vs-conceptual decomposition should be invariant to such reparametrizations. (A4) formalizes PCR as a *share*: equal absolute gains in scaffolded accuracy contribute equally to the recovery score, and a stochastic mixture of two scaffolds receives the convex combination of their individual PCRs, the natural property for a metric reporting the fraction of a gap closed.

Proposition 1 (Uniqueness of PCR)**.** *Any function M satisfying (A1) through (A4) coincides with the Perceptual Contribution Ratio:*

$$M(a,b,c) = \frac{b-a}{c-a}. \tag{S1}$$

Proof. Fix (a,b,c) with $c > a$. By (A2) with $\delta = -a$,

$$M(a,b,c) = M(0,b-a,c-a).$$

Since $c > a$, the scalar $\alpha := (c-a)^{-1}$ is positive, so (A3) applies:

$$M(0,b-a,c-a) = M\left(0, \tfrac{b-a}{c-a}, 1\right) =: g\left(\tfrac{b-a}{c-a}\right),$$

where $g(t) := M(0, t, 1)$. The metric is therefore a univariate function of the affine ratio $t = (b-a)/(c-a)$. By (A1), $g(0) = 0$ and $g(1) = 1$. By (A4), g is affine: $g(t) = pt + q$ for constants $p,q \in \mathbb{R}$. The endpoint conditions force $q = 0$ and $p = 1$, so $g(t) = t$ and

$$M(a,b,c) = \frac{b-a}{c-a}. \qquad \square$$

Necessity of each axiom. Each axiom is independently required. Dropping (A1) admits any affine rescaling $\alpha\mathrm{PCR} + \beta$. Dropping (A2) admits metrics that depend on the absolute origin (e.g., b/c). Dropping (A3) admits metrics that depend on the unit (e.g., the raw gap $b-a$). Dropping (A4) admits any endpoint-preserving reparametrization $M(a,b,c) = g((b-a)/(c-a))$ with $g(0) = 0$ and $g(1) = 1$; on the in-range interval $t \in [0,1]$, examples include t^k for $k > 0$ or $\sqrt{t}$. Such reparametrizations would assign different contribution scores to the same scaffolded accuracy depending on the curvature of the chosen g, with no principled basis for selecting one curvature over another. Among metrics that respect all four invariances, PCR is the unique choice.

Out-of-range values. The proposition makes no assumption that $b \in [a,c]$, so the characterization extends to the empirically observed regime in which $\mathrm{PCR} > 1$ (the scaffold-plus-image combination matches or exceeds the text-only ceiling; §G) and $\mathrm{PCR} < 0$ (scaffold actively interferes with reasoning, observed for the blob-detection parser on Stage 3 and for several mid-scale models on B_a). The same affine ratio governs all three regimes, which is what licenses the dose-response interpretation of the triplet $(\mathrm{PCR}_a, \mathrm{PCR}_p, \mathrm{PCR}_f)$.

E Extended Discussion of Competing Accounts

Several recent works argue that VLM failures on social reasoning tasks are cognitive rather than perceptual. We discuss each in detail and explain why our findings are complementary rather than contradictory.

Liu *et al*. [19]: "Seeing but Not Believing." This work demonstrates that VLMs internally attend to correct visual evidence yet still output incorrect answers on VQA benchmarks. The key distinction from our work lies in the *perceptual extraction complexity* of the tasks. Their benchmarks involve single-image VQA where the relevant visual evidence occupies a spatially localized region within one frame. Our stimuli require *sequential multi-panel entity tracking*. Our data directly address whether this distinction holds: the 60.0% reality-bias error rate shows that models correctly identify where the object currently is but fail to track who was absent when it moved.

Chen *et al*. [8]: "VLMs See What You Want but Not What You See." VLMs achieve high performance on intent recognition but consistently fail at perspective-taking, scoring *below chance* on PerspectBench. Our work offers a complementary interpretation: the egocentric bias they observe is precisely what our reality-bias error category captures (549 of 915 visual-only Stage 3 errors; 60.0%). We demonstrate that this bias disappears when perceptual scaffolding provides explicit agent-perspective information (B_f: 98.4% vs. A: 73.9%).

Tong *et al*. [34]: "Eyes Wide Shut." Their identification of systematic visual shortcomings in multimodal LLMs traceable to CLIP-based encoders provides complementary evidence from the representation-learning side. Our scaffolding intervention demonstrates the downstream consequence: when structured perceptual content is provided externally, the reasoning succeeds, consistent with their finding that visual grounding remains an open challenge.

Zhang *et al*. [39]: "MindPower." Their BDI framework decomposes ToM reasoning into perception, mental reasoning, decision, and action layers, finding that VLMs perform poorly across all layers but especially on mental reasoning. Our work is consistent with their per-layer analysis and adds the *scaffolding intervention* that isolates the perceptual from the conceptual component.

Jin *et al*. [14]: "MMToM-QA." Our benchmark differs in three key respects: (i) Abstract 2D panels rather than naturalistic 3D game scenes; (ii) a scaffolding intervention that disentangles perception from reasoning; (iii) evaluation across a developmental hierarchy rather than testing false belief in isolation.

Shapira *et al*. [28]: "Clever Hans or Neural Theory of Mind?" Our work operates in the visual modality and reaches a structurally different conclusion: for frontier models, the inferential competence (demonstrated by 94.5% text-only accuracy) is available in the text modality, and the visual failure is attributable to perceptual extraction, not pattern matching.

Hu *et al*. [13]: "Re-evaluating Theory of Mind Evaluation in LLMs." Our five-condition protocol directly addresses this concern: by holding the reasoning task constant and varying only the perceptual input modality, we isolate whether the model is performing genuine belief-reality dissociation or relying on linguistic shortcuts.

F Additional Limitations and Caveats

Stimulus and model generalization. Our evaluation covers two stimulus regimes, abstract 2D panels and naturalistic 3D game scenes from MuMA-ToM and MindPower. Generalization to in-the-wild human video, surveillance footage, or first-person egocentric video is not established and is a natural next test. The model lineup is limited to seven systems available through a commercial API during March 2026; ToM-specific fine-tunes, open-weight models with object-centric objectives, and not-yet-deployed multimodal systems are not represented.

Methodological scope. Models are accessed as black boxes through a single inference provider with deterministic decoding; we measure functional consequences on outputs, not attention patterns or internal representations. The MindPower B_f scaffold is an oracle full-information condition rather than a leakage-controlled perceptual ceiling; the perceptual-only finding for MindPower rests on B_p, and the MindPower PCR_p values (Table 2c) are the appropriate quantitative summary on that regime. Reality-bias categorization is based on a deterministic algorithm (§A.5) and inherits its definitions.

Answer balance. All *sally_anne_search* (50 probes) and *sally_anne_belief* (50 probes) have expected answer "left." All *sally_anne_reality* (50 probes) have expected answer "right." Template-reweighted PCR analysis (§G.6) further confirms that conclusions are invariant to answer-balance reweighting.

Abstract 2D panels. The Heider-and-Simmel-style abstraction deliberately isolates cognitive capacity from appearance-based recognition. The perceptual bottleneck identified here is a lower bound: naturalistic 3D game scenes impose greater perceptual demands. Generalization from Abstract 2D panels to naturalistic 3D game scenes is tested separately by MuMA-ToM and MindPower rather than inferred from the panel experiments.

API-based evaluation. All models are accessed through the OpenRouter API as black boxes. Temperature 0 removes stochasticity, meaning each model gives one deterministic answer per probe; error bars reflect cross-probe and cross-model variance, not model uncertainty.

Table S3: B_{vlm} **per-model Stage 3 accuracy** (Stage 3 ablation subset, n=500). Two independent parser models.

Model	B_{vlm}		A	B_p	B_f
	GPT-4o parser	Claude parser			
GPT-4o [23]	81.6	77.2	76.4	90.2	99.2
Gemini 2.5 Pro [10]	79.6	60.8	71.8	90.4	92.6
Claude Sonnet 4 [1]	79.8	76.8	78.8	97.6	100.0
Qwen2.5-VL-72B [3]	62.0	59.6	79.2	90.0	100.0
Llama 4 Mav. [20]	84.0	84.2	87.0	85.0	99.8
GPT-4.1-mini [24]	72.0	74.8	70.2	80.2	100.0
Mean	**76.5**	**72.2**	77.2	88.9	98.6
PCR	−0.04	−0.27	n/a	0.63	1.16

Stage 1 leading prompts. The *count_two* template's answer instruction reads "Answer ONLY '2'," which explicitly contains the expected answer. Within-probe comparisons are unaffected, but this template's absolute Stage 1 accuracy is inflated.

Scaffold fidelity spectrum. The five conditions (A, B_a, B_p, B_f, C) sample the scaffold fidelity continuum at discrete points. The additional $B_{\text{vis-stripped}}$ and B_{vlm} conditions (§G) provide two further data points on this continuum.

Claude text-only Stage 2 anomaly. Claude Sonnet 4 scores 51.5% on text-only goal attribution (Table S1), far below other models (92–100%). Inspection reveals a systematic direction-convention misinterpretation in pure-text probes: Claude interprets "upper" and "lower" differently from the intended spatial convention when no image is present. This anomaly is specific to Stage 2 text-only and does not affect Stage 3 conclusions.

Model API versions. All models were evaluated through OpenRouter. We report the exact model identifiers used:
`openai/gpt-4o`,
`google/gemini-2.5-pro`,
`anthropic/claude-4-sonnet-20250522`,
`qwen/qwen2.5-vl-72b-instruct`,
`meta-llama/llama-4-maverick-17b-128e-instruct`,
`openai/gpt-4.1-mini-2025-04-14`,
`mistralai/ministral-8b-2512`.

G Additional Ablation Experiments

This section stress-tests the perceptual-bottleneck interpretation by asking five failure-mode questions: can a stronger parser rescue the image-only condition, is the gain merely explicit visibility leakage, do text scaffolds survive paraphrase, can the model repair itself with chain-of-thought, and are the results an artifact of template imbalance? The ablation audit contributes 38,500 additional Abstract 2D panel calls.

G.1 Experiment 1: VLM-as-Parser (B_{vlm})

Two frontier VLMs are independently tested as zero-shot visual scene describers. For each of the 500 Stage 3 probes, the parser model is prompted with a system message instructing a panel-by-panel description of entity positions, colors, and inter-panel changes, without any ToM interpretation (max 600 tokens, temperature 0). The resulting narrations are then provided as scaffolds to the ablation subset reported in Table S3.

Table S4: $B_{\text{vis-stripped}}$ **per-model Stage 3 accuracy** (n=500). Compared against B_p and B_f.

Model	$B_{\text{vis-str}}$	B_p	B_f	Δ vs B_f
GPT-4o [23]	99.6	90.2	99.2	(+0.4)
Gemini 2.5 Pro [10]	54.4	90.4	92.6	(−38.2)
Claude Sonnet 4 [1]	100.0	97.6	100.0	0.0
Qwen2.5-VL-72B [3]	99.8	90.0	100.0	(−0.2)
Llama 4 Mav. [20]	98.2	85.0	99.8	(−1.6)
GPT-4.1-mini [24]	88.0	80.2	100.0	(−12.0)
Ministral-8B [21]	90.0	88.4	97.4	(−7.4)
Mean	**90.0**	88.8	98.4	(−8.4)

Neither parser provides an aggregate benefit over visual-only input. GPT-4o-as-parser (76.5%) is statistically indistinguishable from Condition A on the same six-model abstract subset (77.2%; $\text{PCR} = -0.04$); Claude-Sonnet-4-as-parser (72.2%) degrades performance by 5.0 pp ($\text{PCR} = -0.27$). The corresponding values relative to the seven-model main aggregate (Cond A $= 73.9\%$, Cond C $= 94.5\%$) are $\text{PCR}_{\text{vlm}}^{\text{GPT}} = 0.13$ and $\text{PCR}_{\text{vlm}}^{\text{Claude}} = -0.08$. Although both parsers generate substantially richer descriptions than the blob-detection parser, they commit systematic errors: hallucinating entities not present in panels, inconsistently naming colors across panels, and failing to detect exit events. The consistency across two independent frontier parsers confirms that the bottleneck is a fundamental limitation of current visual encoders, not an artifact of any single model's weaknesses.

G.2 Experiment 2: Visibility Ablation ($B_{\text{vis-stripped}}$)

Explicit visibility/absence phrases are removed from B_f scaffolds via regex matching against nine patterns. On average, 67 ± 35 characters (8.9% of scaffold length per probe) are removed per probe, affecting 80% of probes.

Visibility patterns removed:

- Sentences containing: "was/is absent", "did/does not see/observe", "was/is not present", "unaware of", "has no knowledge of", "cannot/could not see", "left the room/scene", "exited the room/scene", "not in the room", "outside the room", "while X was gone/away", "without X knowing/seeing".

The aggregate result ($B_{\text{vis-stripped}} = 90.0\%$; $\text{PCR}_{\text{vis-stripped}} = 0.79$) confirms that most of the scaffold benefit derives from perceptual content rather than explicit visibility predicates. Gemini 2.5 Pro is a notable outlier (54.4%), suggesting model-specific reliance on explicit absence cues; all other models remain at or above 88.0%.

G.3 Experiment 3: Paraphrase Robustness for C and B_f

Three paraphrase variants are applied to both Condition C and B_f scaffolds across the ablation subset and 500 probes:

- **Restructured**: Panel sentences in reverse order.
- **Vocabulary**: Synonym substitution ("character"→"figure", "container"→"box", etc.).
- **Adversarial**: Ullman-style misleading insertion: "Note that all characters could potentially see everything that happens in the scene."

All six paraphrase conditions exceed 93%, confirming that the inferential competence measured by these conditions is robust and not phrasing-dependent.

G.4 Experiment 4: Chain-of-Thought Self-Scaffolding

Each model in this ablation receives a system prompt instructing step-by-step panel description, presence/absence tracking, and then answering (max 600 tokens, temperature 0). The CoT prompt adds no external information.

Table S5: **Paraphrase robustness for C and B_{full}** (Stage 3 ablation subset, n=3,000 per variant).

Condition	Acc (%)	Δ vs original
C (original)	95.7	n/a
$C_{\text{restructured}}$	95.2	(−0.5)
$C_{\text{vocabulary}}$	94.8	(−0.9)
$C_{\text{adversarial}}$	97.2	(+1.5)
B_{full} (original)	98.6	n/a
$B_{f,\text{restructured}}$	93.9	(−4.7)
$B_{f,\text{vocabulary}}$	97.1	(−1.5)
$B_{f,\text{adversarial}}$	97.6	(−1.0)

Table S6: **Chain-of-thought self-scaffolding** (Stage 3 ablation subset, n=500 per model). †Gemini produces truncated markdown reasoning; only 15% of responses were parseable.

Model	A_{CoT}	A	Δ
GPT-4o [23]	76.6	76.4	(+0.2)
Gemini 2.5 Pro† [10]	N/A	71.8	N/A
Claude Sonnet 4 [1]	90.6	78.8	(+11.8)
Qwen2.5-VL-72B [3]	74.0	79.2	(−5.2)
Llama 4 Mav. [20]	62.8	87.0	(−24.2)
GPT-4.1-mini [24]	79.0	70.2	(+8.8)
Mean (excl. Gem.)	**76.6**	78.3	(−1.7)

Table S7: **Self-scaffold generate-then-reason baseline** (Stage 3 ablation subset on Abstract 2D panels). A_{self} asks the model to first describe the panels, then answer using its own description.

Model	A_{self}	A	Δ
GPT-4o [23]	77.6	76.4	(+1.2)
Gemini 2.5 Pro [10]	14.6	71.8	(−57.2)
Claude Sonnet 4 [1]	69.2	78.8	(−9.6)
Qwen2.5-VL-72B [3]	58.6	79.2	(−20.6)
Llama 4 Mav. [20]	56.2	87.0	(−30.8)
GPT-4.1-mini [24]	74.6	70.2	(+4.4)
Mean	**58.5**	77.2	(−18.7)

G.5 Experiment 5: Self-Scaffold Generate-Then-Reason

Each model in this ablation first generates a panel-by-panel scene description (max 600 tokens, temperature 0, no access to the question), then reasons over its own description together with the image (max 200 tokens, temperature 0). Two API calls are used per probe.

Self-scaffolding (A_{self}) yields lower aggregate accuracy than standard Condition A. Only GPT-4o (+1.2 pp) and GPT-4.1-mini (+4.4 pp) show marginal gains; Gemini collapses under the generate-then-reason format, and the other affected models lose accuracy. The result is consistent with the B_{vlm} finding (§G.1): models' own visual descriptions contain too many entity-identity and event-detection errors to serve as effective scaffolds for downstream belief reasoning.

G.6 Template-Reweighted PCR

To address potential inflation from answer-distribution imbalance, we compute PCR after reweighting: each of the 10 Stage 3 templates contributes equally to accuracy regardless of its probe count. Because every template contributes exactly 50 probes in our design, template-reweighting is equivalent to probe-averaging here, so the reweighted values coincide with the main aggregates: $A = 73.9\%$, $B_a = 75.7\%$, $B_p = 88.8\%$, $B_f = 98.4\%$, $C = 94.5\%$, with $\text{PCR}_a^{\text{rew}} = 0.09$, $\text{PCR}_p^{\text{rew}} = 0.73$, $\text{PCR}_f^{\text{rew}} = 1.19$. The same qualitative pattern holds: automatic scaffolding is weak, perceptual scaffolding recovers a large fraction of the deficit, and full scaffolding closes and slightly exceeds the text-only gap.

H Abstract 2D Panels Dataset Overview

The Abstract 2D panels benchmark is designed to evaluate whether VLM failures on visual false-belief tasks are better explained by perceptual bottlenecks or by reasoning bottlenecks. It spans object tracking, goal attribution, and Theory of Mind cases, and it is intentionally organized to mirror the developmental cascade described in the paper: object individuation at 3–5 months, goal attribution at 6–12 months, and false-belief reasoning at roughly 4 years. This developmental structure is part of the benchmark's justification: later false-belief tasks depend on the representational outputs of earlier perceptual achievements, so failures on visual ToM probes may reflect missing perceptual state extraction rather than missing belief reasoning. The benchmark is also intentionally designed to keep the answer space short and tightly controlled so that errors are less confounded by free-form generation. Most targets are therefore simple outputs such as `yes/no`, `left/right`, a number, or a short phrase, with only two genuinely multi-token templates: `unexpected_contents` (`round objects`) and `reality_vs_belief` (`Actually: right. Belief: left.`). The examples below focus on representative cases that make the contrast between image-based and text-only conditions especially easy to inspect.

1. Question Inventory

Stage 1: Object Tracking

Template	Question family	Answer format	Example
`occlusion_consistent`	Motion continuity under occlusion	yes/no	yes
`occlusion_violation`	Motion-continuity violation under occlusion	yes/no	no
`count_two`	Object counting across panels	number	2
`count_three`	Object counting across panels	number	3
`permanence`	Hidden-versus-gone judgment after occlusion	one word	hidden
`color_change`	Identity or color continuity across occlusion	one word	different

Stage 2: Goal Attribution

Template	Question family	Answer format	Example
`goal_upper`	Goal direction toward one of two targets	upper/lower	upper
`detour_obstacle`	Cause of a detour around an obstacle	A/B/C choice	B
`no_goal`	No-goal control: movement not directed to a target	yes/no	no
`wall_removed`	Rational path choice when the obstacle is removed	straight/curved	straight

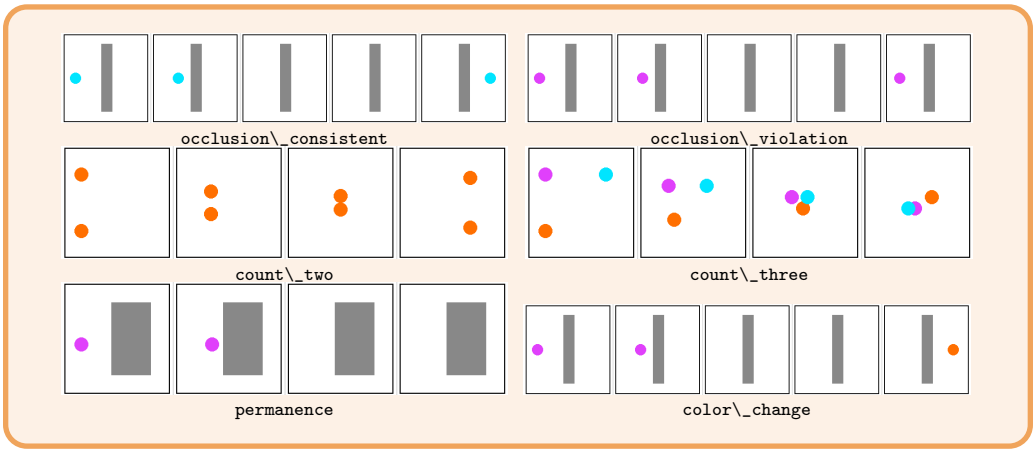
Stage 3: False Belief / Theory of Mind

Template	Question family	Answer format	Example
`sally_anne_search`	False-belief search prediction	left/right	left
`sally_anne_belief`	False-belief location attribution	left/right	left
`sally_anne_reality`	Reality-location check	left/right	right
`reality_vs_belief`	Joint reality-versus-belief dissociation	two-slot	Actual: right; belief: left
`knowledge_who_knows`	Knowledge attribution	stayed/left	stayed
`knowledge_belief`	Belief after an unseen transfer	left/right	left
`unexpected_contents`	Misleading-container contents belief	multi-token	round objects
`two_rooms`	Cross-room partial observability	left/right	right
`perspectives_differ`	Same-versus-different belief comparison	same/different	different
`second_order`	Second-order belief or knowledge	yes/no	yes

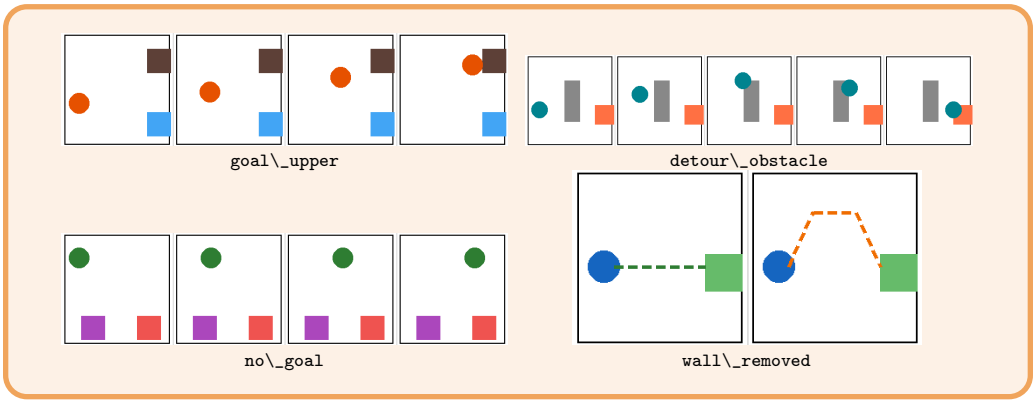
2. Visual Template Grid

Representative strips. The grid below shows one saved panel sequence for each template, organized by developmental stage.

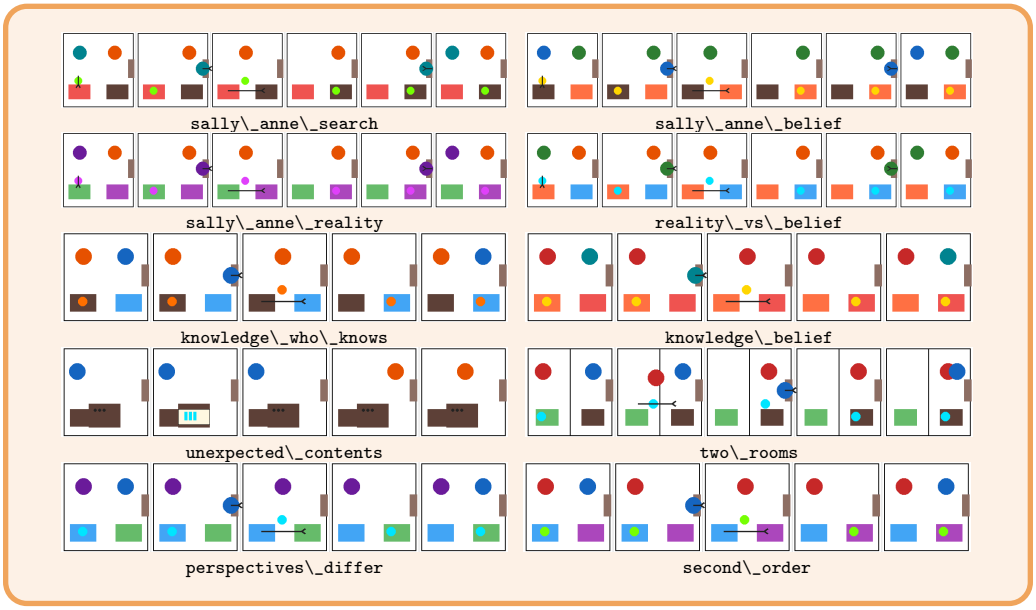
Stage 1: Object Tracking



Stage 2: Goal Attribution



Stage 3: False Belief / Theory of Mind



3. Representative Panel Example for Each Template

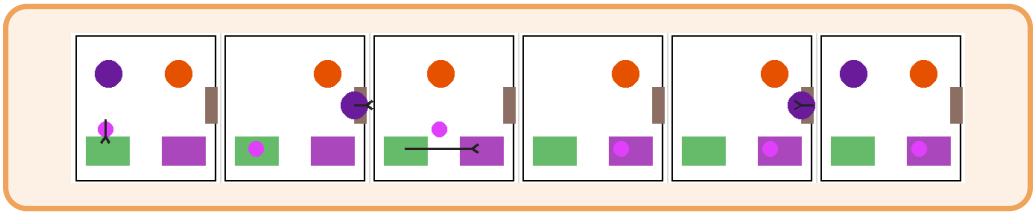
Example outputs. The cases below are sample examples from the saved probes. They use specific saved probe images and report saved responses from `gpt-4o` for conditions `A`, `B_p`, `B_f`, and `C`. The percentages reported below are template-level averages across 350 trials (7 models $\times$ 50 probes), not just the single illustrated case.

`sally\_anne\_reality` **(Stage 3)**

Text-only scene description

```
purple and orange are in a room with a left mint container and a right violet
container.  purple places the pink object in the left container and then leaves.
While purple is away, orange moves the object to the right container.  purple
returns after the move.
```

Visual panels



Why this case is interesting. This is a clean reality-location control where the image-only condition can still fail even though the answer depends only on the object's true final location. In this saved example, `A` fails while `B_p`, `B_f`, and `C` all succeed. Calculated over the full template-level evaluation set, the condition averages are: `A = 74.9%`, `B_p = 98.9%`, `B_f = 99.4%`, `C = 100.0%`.

Model outputs by condition

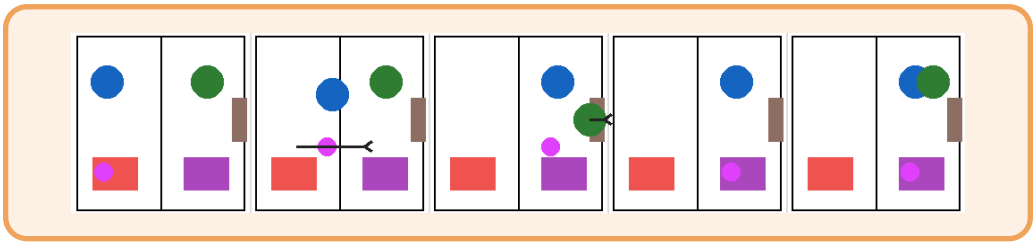
Variant	Setup	Prompt used for model	Model output
A	Image only	Study this 6-panel visual sequence. Where is the pink object actually located in the final panel, left or right? Answer ONLY 'left' or 'right'.	left
B_p	Image + perceptual scaffold	Study this image. Here is a structured perceptual description: Panel 1: purple and orange characters are in the room. pink object is near the left mint container. purple character places the object there. Panel 2: purple character moves to the door and exits. orange character remains in the room. pink object is in the left mint container. Panel 3: Only orange character is in the room. pink object moves from the left mint container toward the right violet container. Panel 4: orange character is in the room. pink object is in the right violet container. Panel 5: purple character returns through the door. orange character is in the room. pink object is in the right violet container. Panel 6: purple and orange characters are both in the room. pink object is in the right violet container. Where is the pink object actually located in the final panel, left or right? Answer ONLY 'left' or 'right'.	right
B_f	Image + full scaffold	Study this image. Here is a full structured description: Panel 1: purple and orange characters are in the room. pink object is near the left mint container. purple character places the object there. Panel 2: purple character moves to the door and exits. orange character remains in the room. pink object is in the left mint container. Panel 3: Only orange character is in the room. pink object moves from the left mint container toward the right violet container. Panel 4: orange character is in the room. pink object is in the right violet container. Panel 5: purple character returns through the door. orange character is in the room. pink object is in the right violet container. purple character was outside when the object moved from left to right and did not see that move. Where is the pink object actually located in the final panel, left or right? Answer ONLY 'left' or 'right'.	right
C	Text only	purple and orange are in a room with a left mint container and a right violet container. purple places the pink object in the left container and then leaves. While purple is away, orange moves the object to the right container. purple returns after the move. Where is the pink object actually located in the final panel, left or right? Answer ONLY 'left' or 'right'.	right

two_rooms (Stage 3)

Text-only scene description

```
blue starts in the left room with the pink object.  green is in the right room.
blue carries the object into the right room while green is watching.  green then
leaves.  While green is gone, blue places the object in the right violet
container.  green returns later.
```

Visual panels



Why this case is interesting. This is a strong single-token failure case: `A = 56.0%`, `B_p = 65.7%`, `B_f = 98.9%`, `C = 78.6%`. In the saved example below, `A` and `B_p` fail, while `B_f` and `C` both succeed.

Model outputs by condition

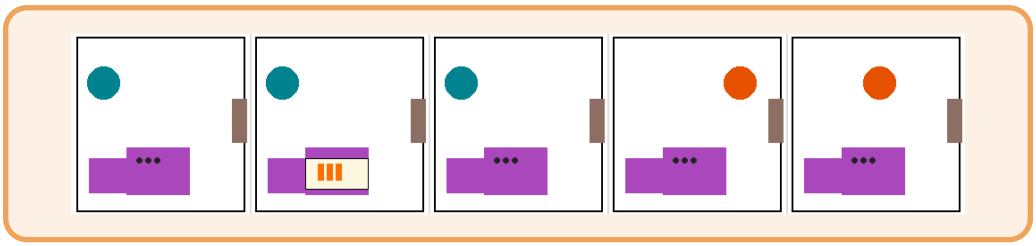
Variant	Setup	Prompt used for model	Model output
`A`	Image only	Study this 5-panel visual sequence. Where does the green character think the pink object is, left or right? Answer ONLY 'left' or 'right'.	`left`
`B\_p`	Image + perceptual scaffold	Study this image. Here is a structured perceptual description: Panel 1: blue character is in the left room with the pink object near the left coral container. green character is in the right room. Panel 2: blue character carries the pink object from the left room toward the right room. green character can see this transfer. Panel 3: green character exits from the right room. blue character and the pink object are in the right room. Panel 4: Only blue character is present. pink object is placed in the right violet container. Panel 5: green character returns to the right room. pink object is in the right violet container. Where does the green character think the pink object is, left or right? Answer ONLY 'left' or 'right'.	`left`
`B\_f`	Image + full scaffold	Study this image. Here is a full structured description: Panel 1: blue character is in the left room with the pink object near the left coral container. green character is in the right room. Panel 2: blue character carries the pink object from the left room toward the right room. green character can see this transfer. Panel 3: green character exits from the right room. blue character and the pink object are in the right room. Panel 4: Only blue character is present. pink object is placed in the right violet container. Panel 5: green character returns to the right room. pink object is in the right violet container. green character left before the object was placed in the right container, but had already seen the object taken into the right room. Where does the green character think the pink object is, left or right? Answer ONLY 'left' or 'right'.	`right`
`C`	Text only	blue starts in the left room with the pink object. green is in the right room. blue carries the object into the right room while green is watching. green then leaves. While green is gone, blue places the object in the right violet container. green returns later. Where does the green character think the pink object is, left or right? Answer ONLY 'left' or 'right'.	`right`

`unexpected\_contents` **(Stage 3)**

Text-only scene description

```
A violet box has round symbols on it.  teal opens it and finds amber rectangular
contents inside.  The box is closed again.  Later, orange arrives and sees only
the closed box with the round symbols.
```

Visual panels



Why this case is interesting. This is one of the two multi-token templates in the benchmark. In this saved example, GPT-4o fails in `A` but succeeds in `B_p`, `B_f`, and `C`, which makes the image-only gap directly visible. Calculated over the full template-level evaluation set, the condition averages are: `A = 55.4%`, `B_p = 100.0%`, `B_f = 99.7%`, `C = 100.0%`.

Model outputs by condition

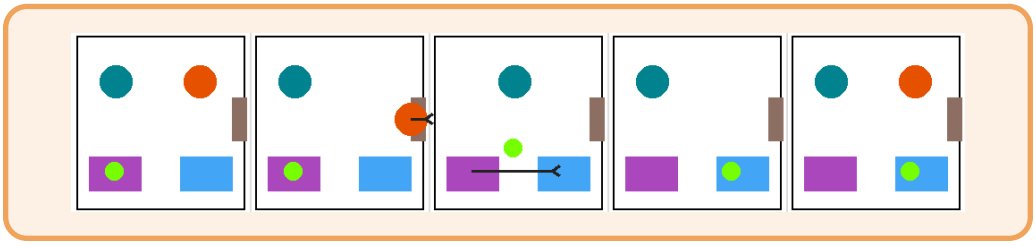
Variant	Setup	Prompt used for model	Model output
`A`	Image only	Study this 5-panel visual sequence. What does the orange character think is inside the box, round objects or yellow rectangles? Answer ONLY 'round objects' or 'yellow rectangles'.	`yellow rectangles`
`B\_p`	Image + perceptual scaffold	Study this image. Here is a structured perceptual description: Panel 1: teal character sees a closed violet box marked with round symbols. Panel 2: teal character opens the violet box. Inside are amber rectangular contents, not round objects. Panel 3: The violet box is closed again and still shows the round symbols. Panel 4: orange character arrives and sees only the closed violet box with round symbols. Panel 5: orange character is still looking at the closed violet box. What does the orange character think is inside the box, round objects or yellow rectangles? Answer ONLY 'round objects' or 'yellow rectangles'.	`round objects`
`B\_f`	Image + full scaffold	Study this image. Here is a full structured description: Panel 1: teal character sees a closed violet box marked with round symbols. Panel 2: teal character opens the violet box. Inside are amber rectangular contents, not round objects. Panel 3: The violet box is closed again and still shows the round symbols. Panel 4: orange character arrives and sees only the closed violet box with round symbols. Panel 5: orange character is still looking at the closed violet box. orange character never saw the box opened and only saw the closed box with the round symbols. What does the orange character think is inside the box, round objects or yellow rectangles? Answer ONLY 'round objects' or 'yellow rectangles'.	`round objects`
`C`	Text only	A violet box has round symbols on it. teal opens it and finds amber rectangular contents inside. The box is closed again. Later, orange arrives and sees only the closed box with the round symbols. What does the orange character think is inside the box, round objects or yellow rectangles? Answer ONLY 'round objects' or 'yellow rectangles'.	`round objects`

`second\_order` (Stage 3)

Text-only scene description

```
teal and orange both see the lime object in the left violet container.  orange
leaves.  While orange is away, teal moves the object to the right light-blue
container.  orange returns after the move.
```

Visual panels



Why this case is interesting. This is a higher-order belief probe: the model must judge whether one agent knows that another agent now has a false belief. In this saved example, A fails while B_p, B_f, and C succeed. Calculated over the full template-level evaluation set, the condition averages are: A = 64.9%, B_p = 97.7%, B_f = 100.0%, C = 90.9%.

Model outputs by condition

Variant	Setup	Prompt used for model	Model output
A	Image only	Study this 5-panel visual sequence. Does the teal character know that the orange character has a wrong belief about the object's location, yes or no? Answer ONLY 'yes' or 'no'.	no
B_p	Image + perceptual scaffold	Study this image. Here is a structured perceptual description: Panel 1: teal and orange characters are in the room. lime object is in the left violet container. Panel 2: orange character exits through the door. teal character remains. lime object is in the left violet container. Panel 3: Only teal character is in the room. lime object moves from the left violet container toward the right light-blue container. Panel 4: teal character is in the room. lime object is in the right light-blue container. Panel 5: orange character returns. teal character is in the room. lime object is in the right light-blue container. Does the teal character know that the orange character has a wrong belief about the object's location, yes or no? Answer ONLY 'yes' or 'no'.	Yes.
B_f	Image + full scaffold	Study this image. Here is a full structured description: Panel 1: teal and orange characters are in the room. lime object is in the left violet container. Panel 2: orange character exits through the door. teal character remains. lime object is in the left violet container. Panel 3: Only teal character is in the room. lime object moves from the left violet container toward the right light-blue container. Panel 4: teal character is in the room. lime object is in the right light-blue container. Panel 5: orange character returns. teal character is in the room. lime object is in the right light-blue container. orange character did not see the move, so orange still has the old belief. teal knows the move happened and knows that orange missed it. Does the teal character know that the orange character has a wrong belief about the object's location, yes or no? Answer ONLY 'yes' or 'no'.	Yes.
C	Text only	teal and orange both see the lime object in the left violet container. orange leaves. While orange is away, teal moves the object to the right light-blue container. orange returns after the move. Does the teal character know that the orange character has a wrong belief about the object's location, yes or no? Answer ONLY 'yes' or 'no'.	yes

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